Grand-Spin $K = 1$ Reduction for the Frozen DTFE Parent Action Explicit Three-Channel Reduction, Symmetry Kernel, Quartic Tensor Derivation, and Reduced Obstruction Theorem

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Abstract

We derive the exact grand-spin $K = 1$ nonradial block of the frozen DTFE parent-action Hessian about the charge-one hedgehog branch. The purpose is narrow, exact, and proof-directed. We do not claim full nonradial closure of the entire parent theory. We isolate the first genuinely coupled nonradial sector beyond the hedgehog block, write it as an explicit three-channel radial variational problem, identify its exact symmetry kernel, derive the complete reduced Euler–Lagrange system, and reduce the remaining non-symmetry question to a single scalar obstruction.

The hedgehog background imported in this reduction is taken as fixed input from the current frozen DTFE configuration used for the $K = 1$ reduction pipeline, and this paper does not itself re-determine or re-normalise the canonical hedgehog shooting value.

The $K = 1$ block is expressed in the true orbital basis ($L = 0, 1, 2$) using the three concrete tensor channels C_0, C_1, C_2. The sigma contribution is written explicitly in covariant Sturm–Liouville form. The quartic Skyrme contribution is then reduced exactly by an adapted-frame tensor decomposition of the linearised current. This produces a fully explicit one-dimensional reduced functional in the three radial channel variables and, after variation, the exact three-channel $K = 1$ ODE system.

Two exact zero modes are identified inside the block: translation and admissible global orientation. These occupy two of the three regular local slots and two of the three admissible decaying far-field slots. After quotienting by this explicit symmetry kernel, the remaining regular class is the $r^2 C_2$ branch and the remaining admissible decaying class is the $r^{-1} C_0$ branch. The full $K = 1$ problem is therefore reduced to a single non-landing obstruction, encoded equivalently by a reduced Evans pairing or by the asymptotic $L = 0$ transversality coefficient of the canonical regular complement solution.

The resulting statement is proof-first in structure. What is established here is the full structural reduction of the $K = 1$ nonradial stability problem to one scalar nondegeneracy target. Numerical or interval-certified evaluation of that scalar remains the final closure step for the $K = 1$ sector.

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1 Introduction

The frozen DTFE matter branch already supports a hedgehog-sector analysis in which existence, strict monotonicity, smoothness, regular origin expansion, quadratic far-field decay, virial balance, non-negativity of the hedgehog Jacobi bottom, and reduction of strict positivity to a sharply isolated nondegeneracy obstruction have been established. The next natural problem is the genuinely nonradial parent-action stability structure.

This document attacks the first new nonradial sector beyond the hedgehog block: grand spin $K = 1$. The aim is not to overstate the status of the full parent theory. The aim is to identify the exact operator in the first coupled nonradial block, isolate its symmetry kernel, derive its reduced three-channel radial formulation, and force the non-symmetry question down to the smallest possible obstruction.

That is the same proof style that made the hedgehog analysis strong: do not diffuse the problem across an undifferentiated basis audit; reduce it to a canonical obstruction with a clean claim boundary.

2 Imported Background from the Frozen Hedgehog Sector

The present reduction imports four background facts from the previously established frozen hedgehog sector.

Proposition 1 (Imported frozen-background facts)**.** *Let $U_*(x)$ be the charge-one hedgehog branch of the frozen parent action. Then the following facts are assumed as established:*

i. There exists a smooth monotone profile $F(r)$ satisfying the hedgehog field equation and the boundary conditions $F(0) = \pi$, $F(\infty) = 0$.

ii. The origin expansion is regular and has the form

$$F(r) = \pi - a_* r + b_* r^3 + \mathcal{O}(r^5) \qquad (r \downarrow 0). \tag{1}$$

iii. The far-field decay is quadratic:

$$F(r) \sim C r^{-2} \qquad (r \to \infty). \tag{2}$$

iv. The purely radial perturbation problem is the grand-spin $K = 0$ block of the full parent-action Hessian.

Only the regular origin structure, the r^{-2} tail, and the prior $K = 0$ identification are needed in what follows.

3 Frozen Parent Action and Hedgehog Background

The frozen DTFE parent action is

$$S[U] = \int \mathrm{d}^4x \left[-\frac{1}{2} \operatorname{Tr}(L_\mu L^\mu) + \frac{1}{16} \operatorname{Tr}\left([L_\mu, L_\nu][L^\mu, L^\nu]\right) \right], \qquad L_\mu := U^\dagger \partial_\mu U, \tag{3}$$

with static energy

$$E[U] = \int_{\mathbb{R}^3} \left[-\frac{1}{2} \operatorname{Tr}(L_i L_i) - \frac{1}{16} \operatorname{Tr}\left([L_i, L_j][L_i, L_j]\right) \right] \mathrm{d}^3x. \tag{4}$$

The charge-one hedgehog branch is

$$U_*(x) = \cos F(r)\, \mathbf{1} + \mathrm{i}\, \hat{x} \cdot \tau \, \sin F(r), \qquad r = |x|, \qquad \hat{x} = \frac{x}{|x|}, \tag{5}$$

with profile equation

$$(r^2 + 2\sin^2 F) F'' + 2rF' + \sin(2F) \left(F'^2 - 1 - \frac{\sin^2 F}{r^2} \right) = 0, \tag{6}$$

subject to

$$F(0) = \pi, \qquad F(\infty) = 0. \tag{7}$$

The hedgehog current is written in body-fixed form as

$$L_i^* = \mathrm{i}\, A_i \cdot \tau, \tag{8}$$

where the coefficient vectors are

$$A_i = F'\, \hat{x}_i\, \hat{x} + \frac{\sin F \cos F}{r} (e_i - \hat{x}_i \hat{x}) + \frac{\sin^2 F}{r} (e_i \times \hat{x}). \tag{9}$$

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The corresponding radial background scalars that repeatedly appear below are

$$\alpha := F'(r), \qquad \beta := \frac{\sin F\cos F}{r}, \qquad \gamma := \frac{\sin^2 F}{r}, \qquad \lambda := \frac{\sin F}{r}, \tag{11}$$

with the exact identity

$$\beta^2 + \gamma^2 = \lambda^2. \tag{12}$$

4 Parent-Action Perturbation and Grand-Spin Decomposition

We perturb at the parent-action level by

$$U_\varepsilon(x) = U_*(x)e^{\varepsilon\Xi(x)}, \qquad \Xi(x) = \mathrm{i}\,\Phi(x)\cdot\tau, \tag{13}$$

with body-fixed adjoint coefficient field $\Phi : \mathbb{R}^3 \to \mathbb{R}^3$.

The first variation of the current is

$$\delta L_i = D_i\Xi, \qquad D_i\Phi = \partial_i\Phi - 2A_i \times \Phi. \tag{14}$$

The hedgehog is invariant under the diagonal action of spatial rotation and matching isorotation, so the Hessian commutes with grand spin

$$\mathbf{K} = \mathbf{L} + \mathbf{I}, \tag{15}$$

with $I = 1$ in the adjoint perturbation representation. Thus the tangent space decomposes into grand-spin blocks. The previously studied hedgehog Jacobi problem is exactly the $K = 0$ block. The first genuinely new nonradial sector is $K = 1$.

5 Concrete $K = 1$ Orbital Basis

Fix a constant vector $a \in \mathbb{R}^3$. Define the three concrete orbital tensors

$$C_0 := a, \qquad C_1 := a \times \hat{x}, \qquad C_2 := 3(a\cdot\hat{x})\hat{x} - a. \tag{16}$$

These are exactly the $L = 0, 1, 2$ channels inside the $K = 1$ grand-spin sector:

$$\Delta_\Omega C_0 = 0, \qquad \Delta_\Omega C_1 = -2C_1, \qquad \Delta_\Omega C_2 = -6C_2. \tag{17}$$

Hence every $K = 1$ perturbation can be written as

$$\Phi_{K=1}[a](r,\Omega) = \psi_0(r)C_0 + \psi_1(r)C_1 + \psi_2(r)C_2. \tag{18}$$

Proposition 2 (Exact angular metric and barrier)**.** *The three channels are orthogonal under angular integration, with*

$$\int_{S^2} |C_0|^2\,\mathrm{d}\Omega = 4\pi|a|^2, \tag{19}$$

$$\int_{S^2} |C_1|^2 \, \mathrm{d}\Omega = \frac{8\pi}{3}|a|^2, \tag{20}$$

$$\int_{S^2} |C_2|^2 \, \mathrm{d}\Omega = 8\pi|a|^2. \tag{21}$$

Thus, after factoring out $4\pi|a|^2$, the channel metric is

$$M = \mathrm{diag}\left(1, \frac{2}{3}, 2\right), \tag{22}$$

and the free angular barrier is

$$G = \mathrm{diag}(0, 2, 6). \tag{23}$$

6 Exact Radial Connection in the $K = 1$ Block

In the (C_0, C_1, C_2) basis, the radial covariant derivative closes exactly:

$$D_r\psi = \psi' + F'(r)\,\mathcal{B}\,\psi, \tag{24}$$

where

$$\psi = \begin{pmatrix} \psi_0 \\ \psi_1 \\ \psi_2 \end{pmatrix}, \qquad \mathcal{B} = \begin{pmatrix} 0 & -\frac{4}{3} & 0 \\ 2 & 0 & -2 \\ 0 & \frac{2}{3} & 0 \end{pmatrix}. \tag{25}$$

Proposition 3 (Metric-skew radial connection)**.** *The connection matrix satisfies*

$$M\mathcal{B} + \mathcal{B}^T M = 0. \tag{26}$$

Consequently the principal radial derivative part of the block is already in covariant Sturm–Liouville form.

7 Exact Symmetry Kernel

There are two exact zero-mode families in the $K = 1$ block for each fixed direction a.

7.1 Translation

Translate the centred hedgehog:

$$U_{*,\varepsilon}(x) = U_*(x - \varepsilon a). \tag{27}$$

Then

$$\delta_a U_* = -a_i \partial_i U_*, \qquad \Xi_{\mathrm{tr},a} = U_*^\dagger \delta_a U_* = -a_i L_i^*. \tag{28}$$

In the (C_0, C_1, C_2) basis, the translation kernel vector is

$$\mathbf{t}(r) = \begin{pmatrix} \frac{1}{3}\left(F'(r) + 2\frac{\sin F(r)\cos F(r)}{r}\right) \\ \frac{\sin^2 F(r)}{r} \\ \frac{1}{3}\left(F'(r) - \frac{\sin F(r)\cos F(r)}{r}\right) \end{pmatrix}. \tag{29}$$

7.2 Admissible global orientation

The admissible global internal symmetry is conjugation

$$U \mapsto AUA^{-1}, \qquad A = e^{\varepsilon T}, \tag{30}$$

which preserves $U(\infty) = 1$. In the (C_0, C_1, C_2) basis the corresponding kernel vector is

$$\mathbf{o}(r) = \begin{pmatrix} \frac{4}{3}\sin^2 F(r) \\ 2\sin F(r)\cos F(r) \\ -\frac{2}{3}\sin^2 F(r) \end{pmatrix}. \tag{31}$$

Both are regular at the origin and decaying at infinity. Their leading behaviours are

$$\mathbf{t}(r) = \begin{pmatrix} -a_* + \mathcal{O}(r^2) \\ \mathcal{O}(r) \\ \mathcal{O}(r^2) \end{pmatrix} \quad (r \downarrow 0), \qquad \mathbf{t}(r) = \begin{pmatrix} \mathcal{O}(r^{-5}) \\ \mathcal{O}(r^{-5}) \\ -Cr^{-3} + \mathcal{O}(r^{-5}) \end{pmatrix} \quad (r \to \infty), \tag{32}$$

$$\mathbf{o}(r) = \begin{pmatrix} \mathcal{O}(r^2) \\ -2a_* r + \mathcal{O}(r^3) \\ \mathcal{O}(r^2) \end{pmatrix} \quad (r \downarrow 0), \qquad \mathbf{o}(r) = \begin{pmatrix} \mathcal{O}(r^{-4}) \\ 2Cr^{-2} + \mathcal{O}(r^{-4}) \\ \mathcal{O}(r^{-4}) \end{pmatrix} \quad (r \to \infty). \tag{33}$$

Proposition 4 (Channel occupancy of the symmetry kernel)**.** *Translation occupies the regular $L = 0$ slot at the origin and the decaying $L = 2$ slot at infinity. Admissible orientation occupies the regular $L = 1$ slot at the origin and the decaying $L = 1$ slot at infinity.*

8 Change of Variables for the Reduced Block

Introduce the combinations

$$S := \psi_0 + 2\psi_2, \qquad D := \psi_0 - \psi_2, \qquad Y := \psi_1, \tag{34}$$

so that

$$\psi_0 = \frac{S + 2D}{3}, \qquad \psi_1 = Y, \qquad \psi_2 = \frac{S - D}{3}. \tag{35}$$

Also define

$$Q := D' - 2\alpha Y, \qquad R := Y' + 2\alpha D. \tag{36}$$

These are the radial covariant combinations that diagonalise the derivative structure of the sigma term.

9 Sigma Contribution in Reduced Form

The sigma contribution to the reduced $K = 1$ functional is

$$Q_{1,\sigma}[\psi] = 4\pi|a|^2 \int_0^\infty \mathcal{L}_\sigma(r;\psi)\,\mathrm{d}r, \tag{37}$$

with exact density

$$\begin{aligned} \mathcal{L}_\sigma = {} & \frac{r^2}{3}\Big(S'^2 + 2Q^2 + 2R^2\Big) + \frac{4}{3}\Big((S-D)^2 + Y^2\Big) \\ & + \frac{16}{3}\Big(\sin^2 F\,(S^2 - SD + D^2 + Y^2) + \sin F\cos F\, SY\Big). \end{aligned} \tag{38}$$

10 Exact Quartic Tensor Derivation

The quartic term is handled by an adapted-frame decomposition of the linearised current. Collect the linearised current coefficients $B_i = (D_i\Phi)$ into the 3×3 matrix $B = (B_i{}^a)_{i,a}$. Relative to the radial/tangential splitting

$$\mathbb{R}^3 = \mathbb{R}\hat{x} \oplus T_{\hat{x}}S^2, \tag{39}$$

define

$$u := \hat{x}^T B\hat{x} \in \mathbb{R}, \tag{40}$$

$$p := \hat{x}^T BP \in T^*_{\hat{x}}S^2, \qquad q := PB\hat{x} \in T_{\hat{x}}S^2, \tag{41}$$

$$C := PBP \in \mathrm{End}(T_{\hat{x}}S^2), \qquad P := I - \hat{x} \otimes \hat{x}. \tag{42}$$

The quartic Skyrme quadratic density around the hedgehog background is exactly

$$\begin{aligned} q_4^{(2)}(B) = {} & 2\lambda^2\nu^2 + \lambda^2(|p|^2 + |q|^2) - 2\alpha\lambda\, q \cdot p + (\alpha^2 + \lambda^2)\,\mathrm{tr}(CC^T) \\ & + 4\alpha\lambda\,\nu\ \mathrm{tr}\, C + 2\lambda^2\Big((\mathrm{tr}\,\mathrm{sym}\, C)^2 - \mathrm{tr}\,\big((\mathrm{sym}\, C)^2\big)\Big). \end{aligned} \tag{43}$$

Because C acts on the two-dimensional tangent plane, the final term may also be written as

$$2\lambda^2\Big((\mathrm{tr}\,\mathrm{sym}\, C)^2 - \mathrm{tr}\,\big((\mathrm{sym}\, C)^2\big)\Big) = 4\lambda^2 \det(\mathrm{sym}\, C). \tag{44}$$

This is the exact invariant tensor form of the quartic Hessian density.

11 Exact Quartic Reduction in the $K = 1$ Basis

For the $K = 1$ field, define the four reduced combinations

$$P := \frac{S - D}{r} + 2\beta Y - 2\gamma D, \qquad H := \frac{Y}{r} + 2\beta D + 2\gamma Y, \tag{45}$$

$$A := \frac{S - D}{r} + 2\gamma S, \qquad B := \frac{Y}{r} + 2\beta S. \tag{46}$$

Then the adapted-frame tensor blocks reduce exactly to

$$\nu = (a \cdot \hat{x})S', \tag{47}$$

$$q = \rho(Qt_1 + Rt_2), \qquad p = \rho(Pt_1 - Ht_2), \qquad \rho := \sqrt{1 - (a \cdot \hat{x})^2}, \tag{48}$$

$$C = (a \cdot \hat{x}) \begin{pmatrix} A & -B \\ B & A \end{pmatrix}. \tag{49}$$

Hence

$$\nu^2 = (a \cdot \hat{x})^2 S'^2, \tag{50}$$

$$|q|^2 = \rho^2(Q^2 + R^2), \qquad |p|^2 = \rho^2(P^2 + H^2), \qquad q \cdot p = \rho^2(QP - RH), \tag{51}$$

$$\mathrm{tr}(CC^T) = 2(a \cdot \hat{x})^2(A^2 + B^2), \qquad \nu\,\mathrm{tr}\, C = 2(a \cdot \hat{x})^2 S'A, \qquad \det(\mathrm{sym}\, C) = (a \cdot \hat{x})^2 A^2. \tag{52}$$

Integrating over the sphere and using

$$\int_{S^2} (a \cdot \hat{x})^2\, \mathrm{d}\Omega = \frac{4\pi}{3}|a|^2, \qquad \int_{S^2} \rho^2\, \mathrm{d}\Omega = \frac{8\pi}{3}|a|^2, \tag{53}$$

one obtains the exact reduced quartic functional

$$Q_{1,4}[\psi] = \frac{8\pi|a|^2}{3}\int_0^\infty \mathcal{L}_4(r;\psi)\,\mathrm{d}r, \tag{54}$$

where

$$\begin{aligned}\mathcal{L}_4 = \lambda^2 S'^2 + \lambda^2(Q^2+R^2+P^2+H^2) - 2\alpha\lambda\,(QP-RH)\\ + (\alpha^2+3\lambda^2)A^2 + (\alpha^2+\lambda^2)B^2 + 4\alpha\lambda\,S'A.\end{aligned} \tag{55}$$

This is the full exact radial reduction of the quartic Skyrme contribution in the $K=1$ block.

12 Exact Reduced $K=1$ Euler–Lagrange System

The full reduced $K=1$ functional is

$$Q_1[\psi] = 4\pi|a|^2\int_0^\infty \left(\mathcal{L}_\sigma + \frac{2}{3}\mathcal{L}_4\right)\,\mathrm{d}r. \tag{56}$$

Define the sigma contributions

$$E_S^\sigma = -\frac{2}{3}(r^2S')' + \frac{8}{3}(S-D) + \frac{16}{3}\Big(\sin^2 F(2S-D) + \sin F\cos F\,Y\Big), \tag{57}$$

$$E_D^\sigma = -\frac{4}{3}(r^2Q)' + \frac{8}{3}\alpha r^2 R - \frac{8}{3}(S-D) + \frac{16}{3}\sin^2 F(-S+2D), \tag{58}$$

$$E_Y^\sigma = -\frac{4}{3}(r^2R)' - \frac{8}{3}\alpha r^2 Q + \frac{8}{3}Y + \frac{32}{3}\sin^2 F\,Y + \frac{16}{3}\sin F\cos F\,S. \tag{59}$$

Define the quartic momenta

$$\Pi_S = 2\lambda^2 S' + 4\alpha\lambda A, \qquad \Pi_D = 2\lambda^2 Q - 2\alpha\lambda P, \qquad \Pi_Y = 2\lambda^2 R + 2\alpha\lambda H. \tag{60}$$

Define the quartic force terms

$$U_S = \frac{2\lambda^2}{r}P - \frac{2\alpha\lambda}{r}Q + 2(\alpha^2+3\lambda^2)\left(\frac{1}{r}+2\gamma\right)A + 4\beta(\alpha^2+\lambda^2)B + 4\alpha\lambda\left(\frac{1}{r}+2\gamma\right)S', \tag{61}$$

$$U_D = 4\alpha\lambda^2 R + 2\lambda^2\left(-\frac{1}{r}-2\gamma\right)P + 4\beta\lambda^2 H + 2\alpha\lambda\left(\frac{1}{r}+2\gamma\right)Q + 4\alpha^2\lambda H + 4\alpha\beta\lambda R - \frac{2(\alpha^2+3\lambda^2)}{r}A - \frac{4\alpha\lambda}{r}S', \tag{62}$$

$$U_Y = -4\alpha\lambda^2 Q + 4\beta\lambda^2 P + 2\lambda^2\left(\frac{1}{r}+2\gamma\right)H + 4\alpha^2\lambda P - 4\alpha\beta\lambda Q + 2\alpha\lambda\left(\frac{1}{r}+2\gamma\right)R + \frac{2(\alpha^2+\lambda^2)}{r}B. \tag{63}$$

Therefore the exact reduced $K=1$ ODE system is

$$E_S^\sigma + \frac{2}{3}(U_S - (\Pi_S)') = 0, \tag{64}$$

$$E_D^\sigma + \frac{2}{3}(U_D - (\Pi_D)') = 0, \tag{65}$$

$$E_Y^\sigma + \frac{2}{3}(U_Y - (\Pi_Y)') = 0. \tag{66}$$

13 Canonical Regular Non-Symmetry Branch

At the origin, the regular Frobenius classes are exactly

$$C_0 + \mathcal{O}(r^2), \qquad rC_1 + \mathcal{O}(r^3), \qquad r^2C_2 + \mathcal{O}(r^4). \tag{67}$$

The first two are occupied by the exact symmetry modes. Therefore the canonical regular non-symmetry solution R_2 is the unique local zero solution satisfying

$$S(r) = 2r^2 + s_4r^4 + \mathcal{O}(r^6), \qquad D(r) = -r^2 + d_4r^4 + \mathcal{O}(r^6), \qquad Y(r) = y_3r^3 + \mathcal{O}(r^5). \tag{68}$$

Equivalently,

$$\psi_2(r) = r^2 + \mathcal{O}(r^4), \qquad \psi_1(r) = \mathcal{O}(r^3), \qquad \psi_0(r) = \mathcal{O}(r^4). \tag{69}$$

14 Exact Obstruction Pairing

For any two zero solutions u, v of the reduced $K = 1$ system, define the conserved covariant Wronskian

$$\mathscr{W}(u, v)(r) = r^2\Big(u^T M D_r v - (D_r u)^T M v\Big). \tag{70}$$

Because the reduced operator is self-adjoint, $\mathscr{W}$ is constant in r.

Let $\mathcal{R}$ be the three-dimensional regular zero-solution space and let $\mathcal{D}$ be the three-dimensional admissible decaying zero-solution space. Then both are Lagrangian for $\mathscr{W}$:

$$\mathscr{W}(u, v) = 0 \qquad \text{for all } u, v \in \mathcal{R}, \tag{71}$$

$$\mathscr{W}(u, v) = 0 \qquad \text{for all } u, v \in \mathcal{D}. \tag{72}$$

The admissible decaying classes at infinity are

$$r^{-1}C_0, \qquad r^{-2}C_1, \qquad r^{-3}C_2. \tag{73}$$

The exact symmetry modes occupy the last two. So after symmetry removal, the only remaining non-symmetry admissible decaying class is the $r^{-1}C_0$ branch.

Let D_0 denote any zero solution with asymptotic normalisation

$$D_0(r) = r^{-1}C_0 + o(r^{-1}) \qquad (r \to \infty). \tag{74}$$

Since R_2 is regular and $\mathbf{t}, \mathbf{o}$ are regular, one has automatically

$$\mathscr{W}(R_2, \mathbf{t}) = 0, \qquad \mathscr{W}(R_2, \mathbf{o}) = 0. \tag{75}$$

Thus the entire non-symmetry obstruction collapses to the single scalar

$$\Theta_{K=1} := \mathscr{W}(R_2, D_0). \tag{76}$$

Theorem 1 (Reduced obstruction theorem)**.** *The canonical regular non-symmetry branch R_2 lands in the remaining decaying $L = 0$ slot if and only if*

$$\Theta_{K=1} = 0. \tag{77}$$

Equivalently, if

$$\Theta_{K=1} \neq 0, \tag{78}$$

then the exact $K = 1$ block admits no additional non-symmetry admissible zero mode, and is therefore coercive modulo the explicit symmetry kernel.

Proof. Let $\mathcal{R}$ denote the three-dimensional regular zero-solution space and let $\mathcal{D}$ denote the three-dimensional admissible decaying zero-solution space for the exact reduced $K = 1$ system. By construction of the conserved covariant Wronskian, both spaces are Lagrangian. The two exact symmetry modes $\mathbf{t}$ and $\mathbf{o}$ belong to both spaces, so their span

$$\mathcal{S} := \operatorname{span}\{\mathbf{t}, \mathbf{o}\} \subset \mathcal{R} \cap \mathcal{D} \tag{79}$$

is a two-dimensional isotropic subspace.

The regular Frobenius classes at the origin are the $L = 0$, $L = 1$, and $L = 2$ classes. The first two are already occupied by translation and admissible orientation, so modulo $\mathcal{S}$ there is exactly one regular line left. By definition, the canonical representative of that quotient line is R_2. Likewise, the admissible decaying classes at infinity are the $r^{-1}C_0$, $r^{-2}C_1$, and $r^{-3}C_2$ classes. The last two are already occupied by the exact symmetry modes, so modulo $\mathcal{S}$ there is exactly one decaying line left, represented by D_0.

Thus the quotient spaces $\mathcal{R}/\mathcal{S}$ and $\mathcal{D}/\mathcal{S}$ are both one-dimensional. The conserved pairing induced by $\mathscr{W}$ descends to these quotient lines, and the scalar $\Theta_{K=1} = \mathscr{W}(R_2, D_0)$ is precisely that induced pairing. If $\Theta_{K=1} = 0$, then the quotient regular line and the quotient decaying line coincide, so there exists a non-symmetry zero mode that is both regular and admissibly decaying. Conversely, if such a non-symmetry zero mode exists, then it belongs to both quotient lines, forcing their induced pairing to vanish, hence $\Theta_{K=1} = 0$.

Therefore $\Theta_{K=1} = 0$ is equivalent to landing of the regular complement in the remaining decaying $L = 0$ slot, while $\Theta_{K=1} \neq 0$ excludes that landing and yields coercivity modulo the explicit symmetry kernel. □

Remark 1. The theorem shows that the full $K = 1$ nonradial problem has been reduced to a single nondegeneracy target for the exact reduced system. No broader spectral claim is being made here. The only remaining closure step for the $K = 1$ sector is to prove that $\Theta_{K=1}$ does not vanish.

15 Asymptotic Transversality Form

If the channel-$L = 0$ component of R_2 has asymptotic form

$$\psi_0^{(R_2)}(r) = A_0 + \frac{B_0}{r} + o(r^{-1}), \tag{80}$$

then, with D_0 normalised by $D_{0,0}(r) \sim r^{-1}$, one obtains

$$\Theta_{K=1} = -\lim_{R\to\infty}\left(R\frac{\mathrm{d}}{\mathrm{d}R}\psi_0^{(R_2)}(R) + \psi_0^{(R_2)}(R)\right) = -A_0. \tag{81}$$

So the reduced obstruction may be read equally as the constant asymptotic coefficient of the $L = 0$ channel of the canonical regular complement.

16 Claim Boundary and Remaining Task

What has been established exactly is:

1. the concrete orbital basis for the $K = 1$ sector,
2. the exact radial covariant connection,

3. the exact sigma contribution,
4. the exact quartic tensor derivation and its complete radial reduction,
5. the exact three-channel reduced ODE system,
6. the exact symmetry kernel,
7. the canonical regular non-symmetry branch,
8. the reduction of the entire $K = 1$ question to the single scalar obstruction $\Theta_{K=1}$.

What is not claimed here is an interval-certified proof of the nonvanishing of $\Theta_{K=1}$. That final nondegeneracy step requires either a fully certified outward solve of the exact reduced system or a new analytic argument that forces the asymptotic $L = 0$ constant coefficient of R_2 to be nonzero.

17 Conclusion

The first genuinely coupled nonradial block of the frozen DTFE parent-action Hessian has now been reduced to its exact proof object. The $K = 1$ sector is an explicit three-channel radial system in the true orbital basis ($L = 0, 1, 2$). Its sigma contribution is exact. Its quartic Skyrme contribution has been reduced exactly to a one-dimensional variational density in the three channel variables. Its two exact symmetry modes, translation and admissible global orientation, are identified explicitly. The unique regular non-symmetry branch is isolated. The full $K = 1$ obstruction is then reduced to one scalar,

$$\Theta_{K=1} = \mathscr{W}(R_2, D_0), \tag{82}$$

which vanishes if and only if the regular complement lands in the remaining decaying $L = 0$ slot.

So the infinite-dimensional nonradial question has been forced down to a finite, canonical, and sharply bounded target. The remaining task is no longer discovery of the correct operator or kernel structure. It is only the final nondegeneracy closure of the exact reduced $K = 1$ obstruction.